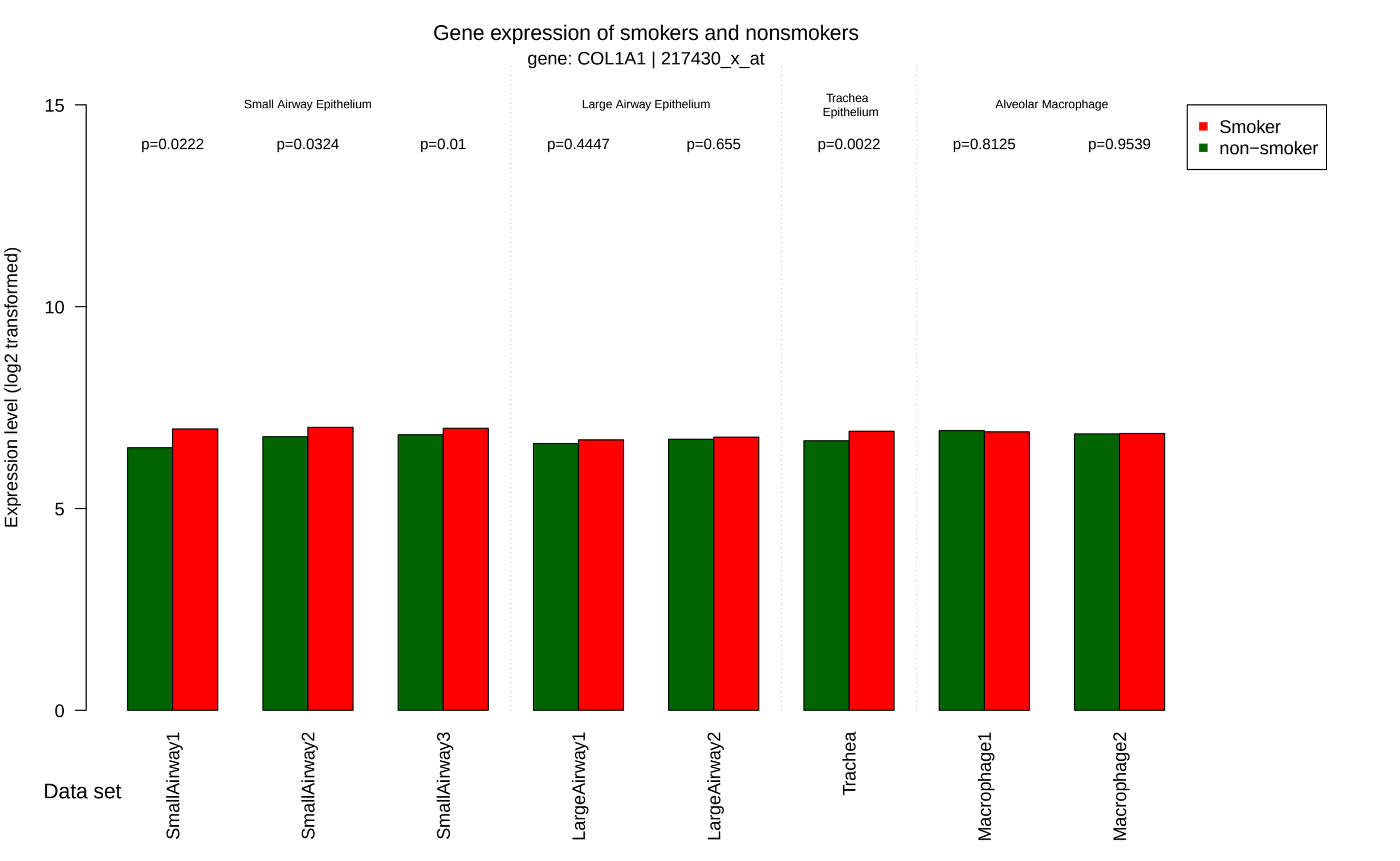
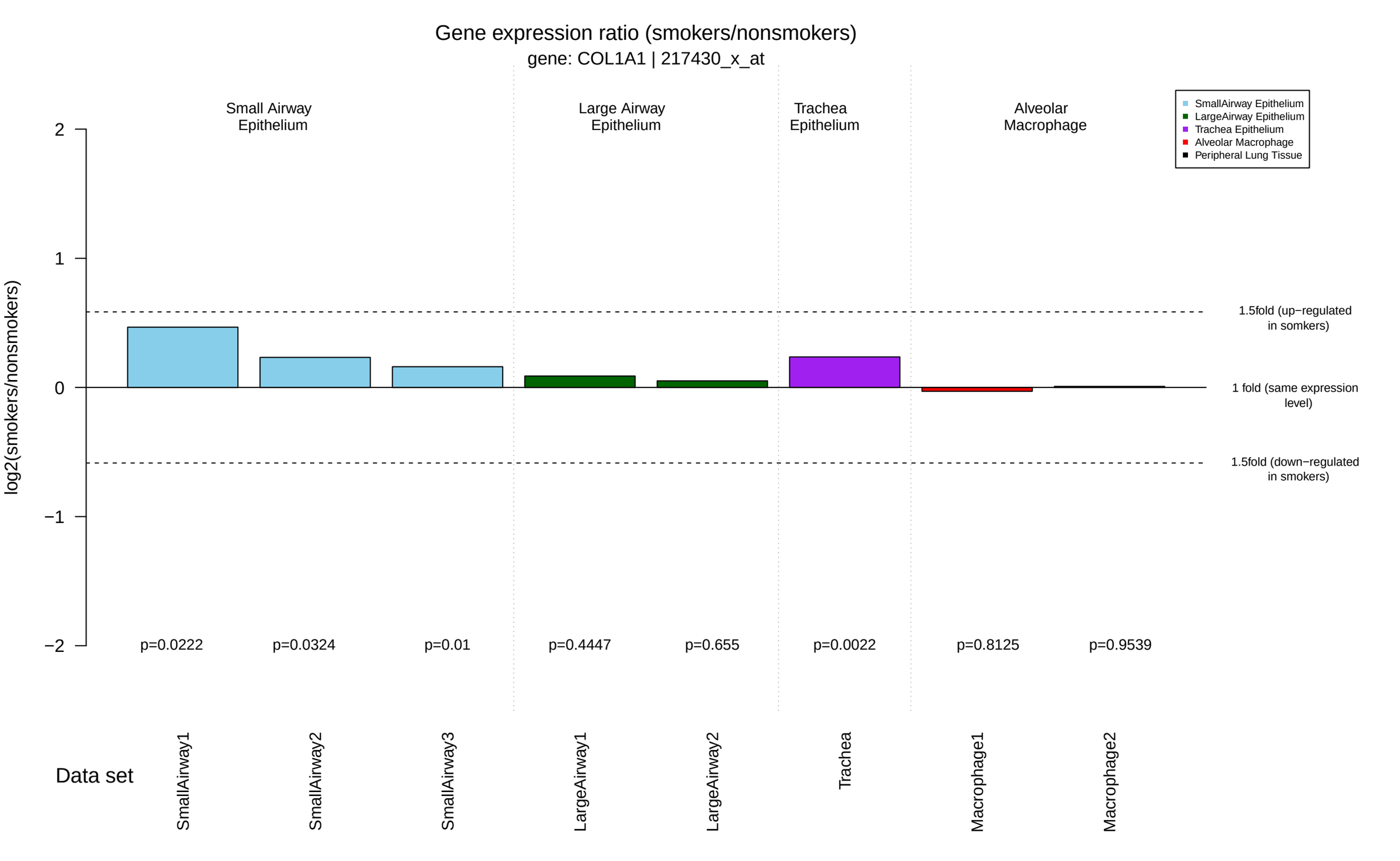
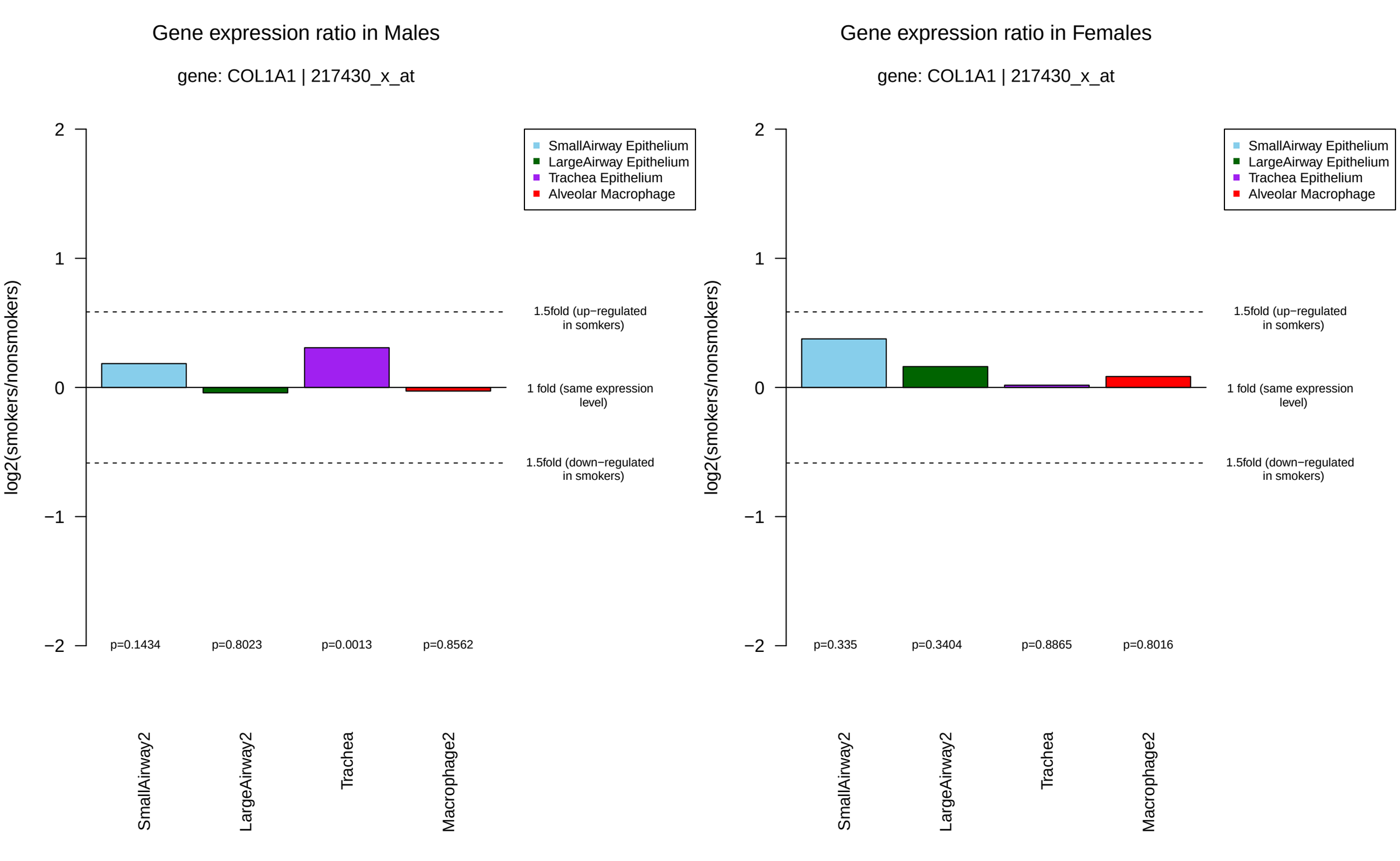
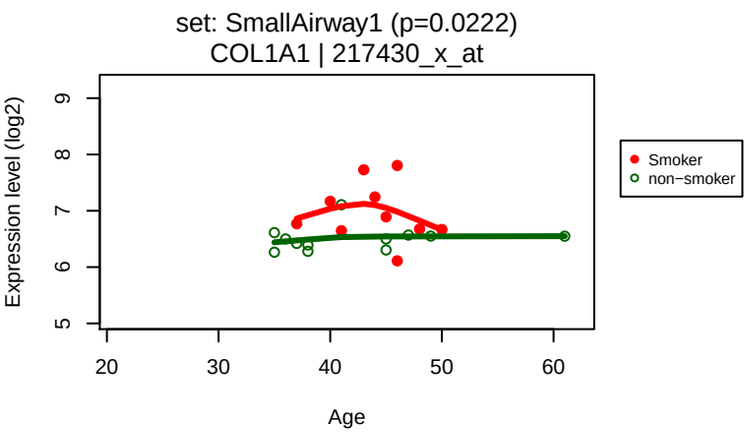
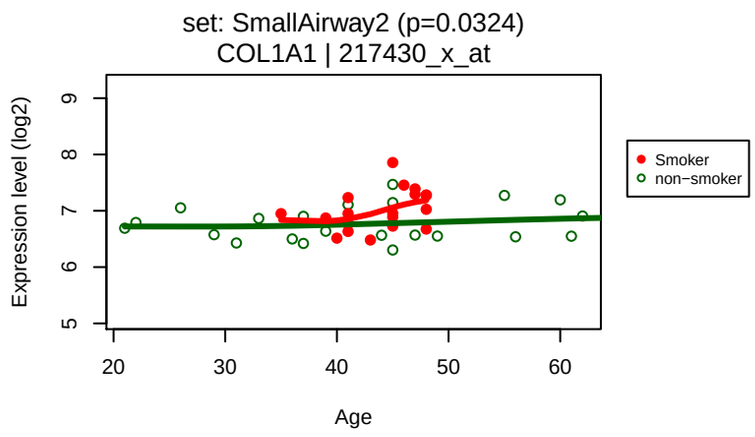
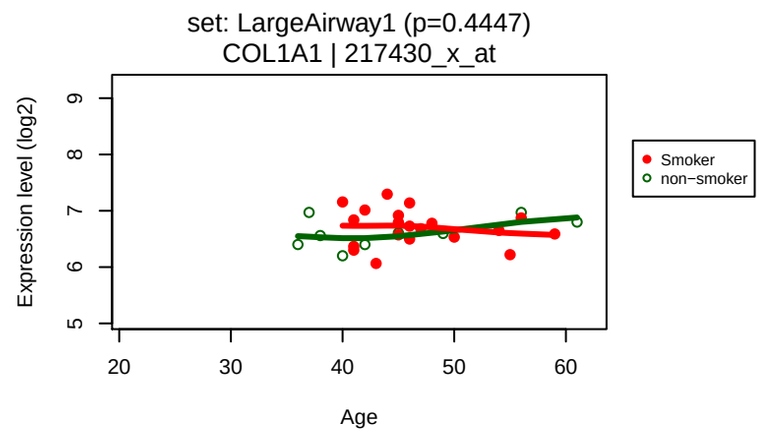

Gene expression of smokers and nonsmokers
gene: COL1A1 | 217430_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0222
p=0.0324
p=0.01
p=0.4447
p=0.655
p=0.0022
p=0.8125
p=0.9539
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

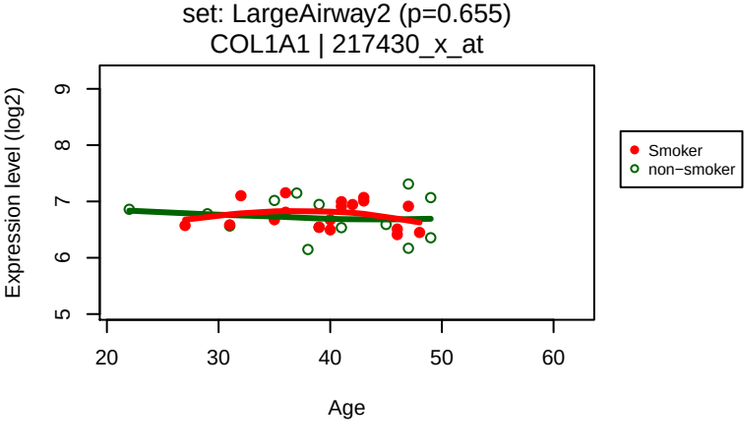

Gene expression ratio (smokers/nonsmokers)
gene: COL1A1 | 217430_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0222
p=0.0324
p=0.01
p=0.4447
p=0.655
p=0.0022
p=0.8125
p=0.9539
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

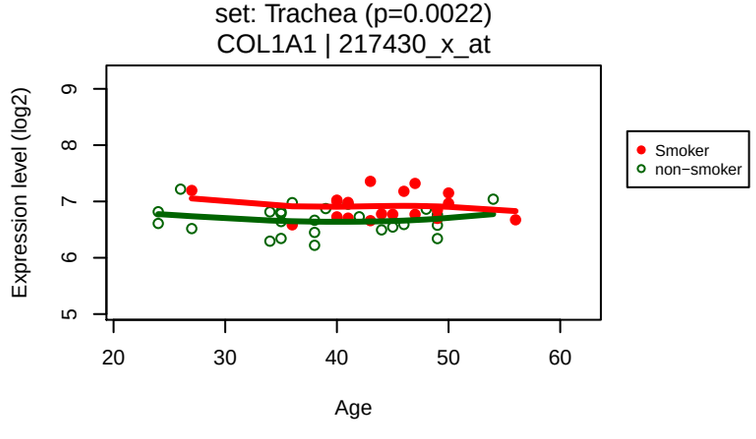

Gene expression ratio in Males
gene: COL1A1 | 217430_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1434
p=0.8023
p=0.0013
p=0.8562
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL1A1 | 217430_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.335
p=0.3404
p=0.8865
p=0.8016
SmallAirway2
LargeAirway2
Trachea
Macrophage2

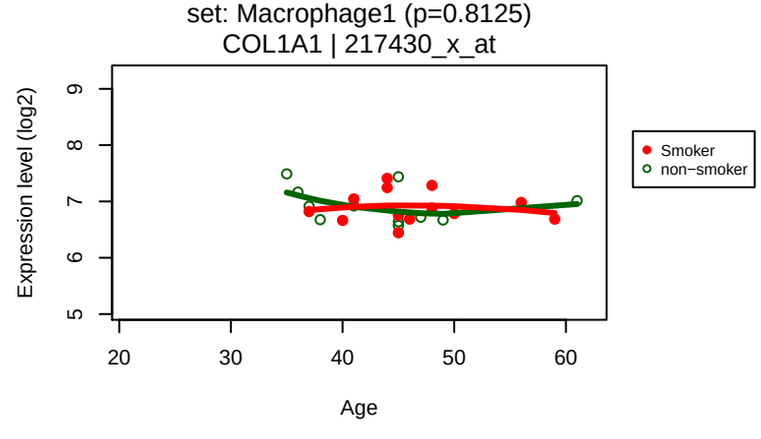

set: SmallAirway1 (p=0.0222)
COL1A1 | 217430_x_at
Expression level (log2)
Age
Smoker
non−smoker

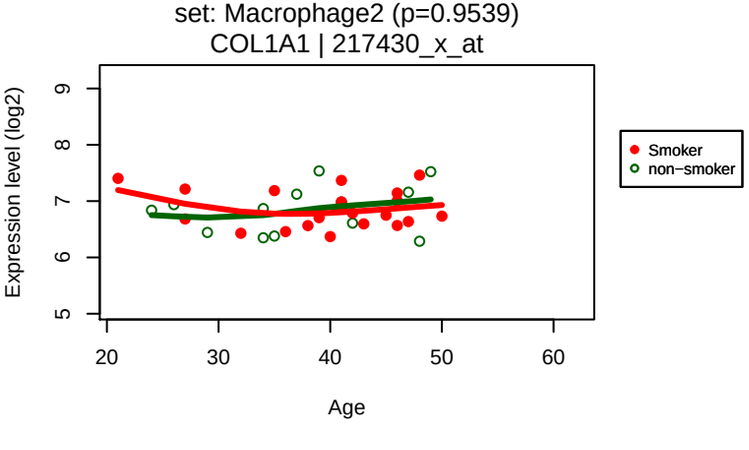

set: SmallAirway2 (p=0.0324)
COL1A1 | 217430_x_at
Expression level (log2)
Age
Smoker
non−smoker

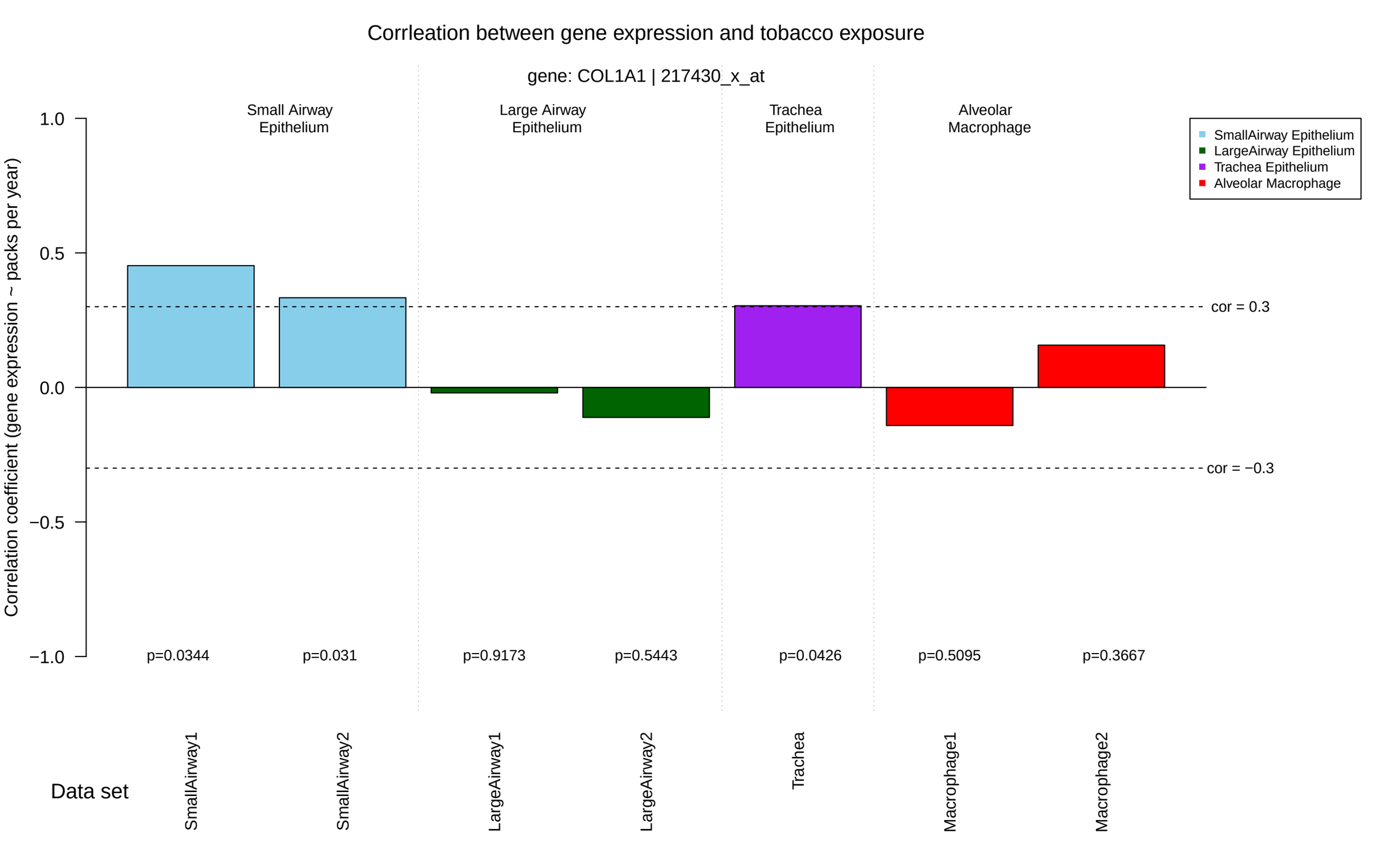

set: LargeAirway1 (p=0.4447)
COL1A1 | 217430_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.655)
COL1A1 | 217430_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0022)
COL1A1 | 217430_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8125)
COL1A1 | 217430_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9539)
COL1A1 | 217430_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL1A1 | 217430_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0344
p=0.031
p=0.9173
p=0.5443
p=0.0426
p=0.5095
p=0.3667
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2